Track 3: Drones and Unmanned Aerial Vehicle (UAV) Technology

This track explores the use of unmanned aerial vehicles for observing, mapping, and treating farmland. Participants should consider regulatory requirements, flight safety, and the practical limits of aerial observation, and should present aerial findings as indications that support, not replace, ground verification.

Problem Title 1: Drone-Based Crop Health Monitoring

A. Problem Description

Walking across large fields to inspect crops is slow, and problems in the middle of a field may go unnoticed until they spread. Crop stress caused by water shortage, nutrient deficiency, or disease may appear as changes in colour or canopy density that are easier to see from above.

Aerial imagery can cover large areas quickly and highlight zones that deserve closer inspection.

B. Challenges

1. Variation in lighting, shadows, and weather during image capture.
2. Limited flight time, payload, and coverage per battery charge.
3. Interpreting visible changes correctly, since several causes can produce similar symptoms.
4. Processing large image sets efficiently and presenting results simply.
5. Complying with applicable drone regulations and operating safely.

Problem Title 2: Precision Spraying and Agricultural Chemical Optimization

A. Problem Description

Blanket spraying across an entire field can waste chemicals, raise costs, and expose workers and the environment to unnecessary risk. Manual spraying is also physically demanding and hazardous to the operator.

Drone-assisted spraying directed at selected zones can reduce chemical use and operator exposure when carried out with proper planning and safeguards.

B. Challenges

1. Controlling droplet size, spray pattern, and drift under changing wind conditions.
2. Limited payload and battery capacity, which restrict area covered per flight.
3. Accurate navigation and spray control over uneven terrain and near obstacles.
4. Determining suitable application rates and ensuring compliance with safety and regulatory guidelines.
5. Safe handling of chemicals and cleaning of equipment.

Problem Title 3: Drone-Based Agricultural Mapping and Land Surveying

A. Problem Description

Many farmers lack up-to-date maps of their land showing boundaries, drainage paths, slopes, and uneven areas. Traditional surveying is time-consuming and costly, and the absence of accurate maps complicates irrigation planning, drainage design, and land-use decisions.

Drone imagery can be stitched into orthomosaic maps and elevation models that provide a consistent spatial reference.

B. Challenges

1. Planning flight paths with adequate image overlap for reliable stitching.
2. Achieving positional accuracy without costly ground-control equipment.
3. Processing large datasets within limited computing resources.
4. Handling tree cover and crop canopy that obscure ground features.
5. Presenting maps in formats that non-specialists can read and use.

Problem Title 4: UAV-Based Pest Infestation Hotspot Detection

A. Problem Description

Pest outbreaks often begin in small patches before spreading across a field. Scouting entire fields on foot is slow, so early patches may be missed. Aerial images can reveal patterns of damage, such as thinning canopy or discoloured patches, that suggest where to look.

It is important to note that such patterns are only indicators. Aerial observation may point to possible infestation zones but cannot independently confirm the exact pest species, which requires ground inspection.

B. Challenges

1. Separating pest damage from similar patterns caused by drought, nutrient deficiency, or disease.
2. Detecting early, small-scale damage at practical flight altitudes and image resolutions.
3. Obtaining labelled data that links aerial patterns to confirmed ground findings.
4. Managing false positives and false negatives and communicating uncertainty.
5. Integrating hotspot maps into a simple scouting workflow.